RESEARCH ARTICLE

Evaluating Prehospital Stroke Scales for Large Vessel Occlusion

A Systematic Review and Network Meta-Analysis

Noah Lee Ahmad Nawabi,[1,2,*] Esteban Rivera Rivera,[3,*] Daniel de Wilde,[1] John L. Kilgallon,[1,4] David I. Nawabi,[1] Varun M. Bhave,[5] Patrick Emedom-Nnamdi,[6] Ari Kappel,[7] Shivani D. Rangwala,[8] Rodolfo E. Alcedo Guardia,[9] Juan Vicenty-Padilla,[9] Saef Izzy,[10] Nirav J. Patel,[7] Rose Du,[7] Adam A. Dmytriw,[11] Toby Gropen,[12] David S. Liebeskind,[13] Edoardo Gaude,[14] Alfred Pokmeng See,[8] Mohammed Ali Aziz-Sultan,[7] and Joshua D. Bernstock[7]

Correspondence
Dr. Bernstock
jbernstock@bwh.harvard.edu

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Abstract

Background and Objectives

Ischemic stroke remains a leading cause of death and disability worldwide, with large vessel occlusion (LVO) accounting for a disproportionate share of poststroke morbidity. Early identification of LVO is essential for timely intervention with endovascular thrombectomy; however, the clinical scales currently used for triage vary widely in their application and accuracy. This study assesses the diagnostic performance of clinical stroke scales in predicting LVO.

Methods

A systematic review was conducted to identify studies evaluating the diagnostic accuracy of prehospital stroke scales for detecting LVO. Pooled sensitivity and specificity were estimated using a bivariate random-effects model, with diagnostic performance further assessed through summary receiver operating characteristic (ROC) curves and area under the curve (AUC) analysis. A Bayesian network meta-analysis was conducted to rank the scales using surface under the cumulative ranking (SUCRA) probabilities, and post hoc analyses were performed to evaluate publication bias.

Results

A total of 58 studies comprising 58,381 patients and 33 unique stroke scales were included in the final analysis. The studies, published between 2014 and 2023, were primarily conducted in North America (50%) and Europe (26%), with a median sample size of 473 participants. Pooled sensitivity ranged from 0.30 (HEMIPARESIS) to 0.99 (LARIO) while specificity varied from 0.34 (FANG) to 0.94 (HEMIPLEGIA). Among the highest-performing scales overall were LARIO (AUC = 0.983), FPSS (AUC = 0.896), FACE2AD (AUC = 0.876), and ACT-FAST (AUC = 0.873). In prehospital settings, FPSS (AUC = 0.896), FAST VAN (AUC = 0.878), and FACE2AD (AUC = 0.876) demonstrated strong performance while LARIO (AUC = 0.983) and ACT-FAST (AUC = 0.883) showed the highest accuracy in hospital settings. Bayesian network meta-analysis identified POMONA (SUCRA = 0.877), NIHSS (0.856), sNIHSS EMS (0.854), G-FAST (0.823), and SAFE (0.788) as the top-ranked scales. Funnel plot analysis revealed minimal publication bias among the most frequently evaluated tools, including RACE, CPSS, and NIHSS.

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*These authors are co-first authors.

[1]Computational Neuroscience Outcomes Center, Department of Neurosurgery, Brigham and Women's Hospital, Harvard Medical School, Boston, MA; [2]College of Medicine, Medical University of South Carolina, Charleston, SC; [3]School of Medicine, University of Puerto Rico, Medical Sciences Campus, San Juan, PR; [4]Hackensack Meridian School of Medicine, Nutley, NJ; [5]Harvard Medical School, Boston, MA; [6]Department of Biostatistics, Harvard T.H. Chan School of Public Health, Boston, MA; [7]Department of Neurosurgery, Brigham and Women's Hospital, Boston, MA; [8]Department of Neurosurgery, Boston Children's Hospital, MA; [9]Neurosurgery Section, School of Medicine, University of Puerto Rico, Medical Sciences Campus, San Juan, PR; [10]Department of Neurology, Brigham and Women's Hospital, Boston, MA; [11]Neuroendovascular Program, Massachusetts General Hospital, Harvard Medical School, Boston, MA; [12]Department of Neurology, The University of Alabama at Birmingham Heersink School of Medicine, AL; [13]Comprehensive Stroke Center and Department of Neurology, University of California, Los Angeles, CA; and [14]Upfront Diagnostics, University of Cambridge, CRUK Cambridge Institute, United Kingdom.

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Glossary

AUC = area under the curve; **CPSSS** = Cincinnati Prehospital Stroke Severity Scale; **EMS** = emergency medical services; **EMSA** = Emergency Medical Stroke Assessment; **EVT** = endovascular thrombectomy; **FAST** = Face, Arm, Speech, Time; **FAST-ED** = Field Assessment Stroke Triage for Emergency Destination; **GFAP** = glial fibrillary acidic protein; **LVO** = large vessel occlusion; **NMA** = Network Meta-Analysis; **PRISMA** = Preferred Reporting Items for Systematic Reviews and Meta-Analyses; **RACE** = Rapid Arterial Occlusion Evaluation; **ROC** = receiver operating characteristic; **SUCRA** = surface under the cumulative ranking; **TCS** = thrombectomy-capable stroke center.

Discussion

Numerous clinical scales are available for detecting LVO in the prehospital setting. While several demonstrate strong performance in specific contexts, there remains a clear need for a simple, accurate, and generalizable tool to reliably identify patients with LVO across diverse clinical environments.

Introduction

Ischemic stroke is the second leading cause of death and the third leading cause of disability worldwide,[1] with global economic consequences estimated at over $960 billion USD in 2019.[2] Large vessel occlusion (LVO), accounting for approximately one-third of all ischemic strokes, is estimated to cause up to 96% of poststroke disability, thereby contributing disproportionately to the overall burden of ischemic stroke.[3] The management of LVO has advanced significantly over the past decade, driven by the emergence and widespread adoption of endovascular thrombectomy (EVT) as the standard of care.[4-7]

Effective treatment of patients with LVO depends heavily on rapid identification and timely transport to thrombectomy-capable stroke centers (TCSs), including comprehensive stroke centers (CSCs) equipped to perform EVT and provide specialized stroke care.[5,8] However, a significant proportion of patients ultimately treated at these centers initially present either by emergency medical services (EMS) or through transfers from outside emergency departments. In addition, multiple studies have underscored the potential benefits of direct transfer to TSCs and the drawbacks of delayed transfer from non–thrombectomy-capable centers.[9-12] Therefore, the ability of EMS personnel in the field and clinicians at smaller, non–thrombectomy-capable facilities to promptly and accurately identify patients who require transfer to a TCS is critical to optimize stroke outcomes.[13]

As a result, increasing emphasis has been placed on the development and validation of clinical stroke scales to aid EMS providers and physicians in identifying patients who may require urgent transfer to TCSs. These tools are designed to stratify the likelihood of LVO based on bedside neurologic assessments and are intended for use in prehospital or early in-hospital settings. To be clinically valuable, such scales must demonstrate high sensitivity (which allows for appropriate identification of patients with LVO) and specificity (which allows for appropriate allocation of resources). Compromising either can lead to inefficient use of bandwidth through unnecessary transfers or, more critically, missed opportunities for timely intervention, with potentially devastating consequences for patients.[14] Scales such as Face, Arm, Speech, Time (FAST); Field Assessment Stroke Triage for Emergency Destination (FAST-ED)[15]; Rapid Arterial Occlusion Evaluation (RACE)[16]; Emergency Medical Stroke Assessment (EMSA)[17]; and the Cincinnati Prehospital Stroke Severity Scale (CPSSS)[18] have become some of the most commonly used tools in prehospital stroke evaluation. However, these instruments often rely on subjective clinical interpretation and are susceptible to various biases that may compromise their diagnostic accuracy and overall reliability.[14,19,20] As the number of available scales continues to grow, it has become increasingly difficult to determine which are best suited for reliably identifying LVO and guiding timely transfer to centers capable of performing EVT.

To address this challenge, this study conducts a comprehensive systematic review and meta-analysis of the current literature assessing the diagnostic performance of prehospital clinical stroke scales in predicting LVO.

Methods

Search Strategy

A systematic literature search was conducted across PubMed, MEDLINE, Web of Science, Embase, and the Cochrane Library to identify studies evaluating the diagnostic performance of prehospital clinical stroke scales for the detection of LVO. This study was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

Study Selection

Study screening and data extraction were performed independently by at least 2 reviewers, following the PRISMA 2020 guidelines. Any discrepancies were resolved through discussion or consultation with a third reviewer. Studies were included if they met the following criteria: (1) reported data

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on the sensitivity, specificity, and/or accuracy of clinical stroke scales; (2) evaluated these outcomes specifically in the context of LVO detection; and (3) described cohorts of patients who had these scales applied during their clinical care. Studies were excluded if they (1) assessed stroke scales solely for non-LVO conditions such as transient ischemic attacks or hemorrhagic strokes; (2) lacked statistical data on sensitivity, specificity, or accuracy; (3) applied stroke scales retrospectively using electronic medical record review; or (4) were abstracts only, systematic reviews or meta-analyses, or published in languages other than English.

Data Extraction

From each included study, the following data were extracted: study characteristics, including design, country, year, and duration of the study; inclusion and exclusion criteria (e.g., symptoms, age, and last-known-well cutoffs); and any reported funding sources. Stroke scale–specific information was also collected, including the index test assessed and its cutoff value (e.g., CPSS score >1), the setting in which it was administered (e.g., ambulance and emergency department), the personnel involved in administration (e.g., EMS providers, ED staff, and stroke team), and the reference standard used to confirm the presence of LVO (e.g., computed tomography angiography and MRI). Finally, diagnostic performance data were extracted, including sample size, number of patients excluded from analysis, sensitivity, specificity, accuracy, positive predictive value, negative predictive value, and reported variance or confidence intervals for each metric.

Statistical Analysis

The primary outcomes of this study were the pooled sensitivity and specificity of each included stroke scale for the identification of LVO. Because LVOs represent approximately 30%–40% of all ischemic strokes, yet account for over 95% of poststroke mortality, identifying tools with high sensitivity and specificity for LVOs specifically was prioritized.[3,21] All statistical analyses were conducted using RStudio (Version 2024.12.0, Build 467; RStudio PBC, Massachusetts). Owing to anticipated heterogeneity among studies, random-effects models were used throughout the analysis.

To assess the diagnostic performance of prehospital stroke scales, 2 complementary meta-analytic approaches were applied. First, a pairwise meta-analysis using a bivariate random-effects model was performed to pool sensitivity and specificity estimates across studies. All pooled effect sizes were reported with corresponding 95% confidence intervals. Summary receiver operating characteristic (sROC) curves and area under the curve (AUC) values were also calculated for each index test to provide a comprehensive measure of diagnostic accuracy.

To enable a systematic comparison of accuracy across all stroke scales, a Bayesian network meta-analysis (NMA) was conducted using the "gemtc" package in RStudio. This methodology has been used extensively in the literature to assess the comparative performance of diagnostic tools in the setting of pulmonary embolism identification and cancer diagnostics, among others.[22-24] In this context, the NMA considered each stroke scale and its respective threshold as separate entities, whereas the bivariate meta-analysis was conducted independently of the threshold at which the stroke scales were evaluated. Stroke scales without a specified threshold score were excluded from the NMA. The network meta-analysis incorporated both direct and indirect comparisons. A network diagram was generated to visualize the relationships between the included stroke scales, with each node representing a scale and its diagnostic threshold. Node size reflected the number of studies evaluating each scale while edge thickness represented the number of direct comparisons. Only studies with comparisons were considered for the network graph. Diagnostic performance was ranked using surface under the cumulative ranking (SUCRA) probabilities to estimate the relative effectiveness of each tool for detecting LVO.

A post hoc analysis was performed for stroke scales evaluated in 5 or more studies, using funnel plots to assess potential publication bias. Statistical significance was defined as $p < 0.05$. All graphical outputs were generated using Excel (Version 2408, Microsoft, Washington) and RStudio.

Ethics Approval and Data Availability

Institutional review board approval was not required for this retrospective, observational literature review because no patient enrollment was involved. Data not provided in the article because of space limitations may be shared at the request of any qualified investigator for the purposes of replicating procedures and results.

Results

Systematic Review

A total of 26,352 records were identified through the initial database search. After deduplication, 9,331 unique studies remained. After title and abstract screening, 516 full-text articles were assessed for eligibility. Of these, 269 studies underwent detailed screening. Among the 269 full-text articles, exclusions were made for the following reasons: 77 reported outcomes that did not meet inclusion criteria, 47 evaluated interventions outside the scope of this review, 33 used ineligible study designs, 18 assessed inappropriate patient populations, 13 provided insufficient data for analysis, 12 were non–English language publications, and 11 addressed unrelated clinical indications. Note that reasons for exclusion were not mutually exclusive. Ultimately, 58 studies met all inclusion criteria and were included in the final analysis. A detailed overview of the screening process is demonstrated in eFigure 1.

Study Demographics and Clinical Characteristics

A total of 58 studies, involving 58,381 participants, were included in the quantitative analysis. Notably, 2 studies were

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Table 1 General Characteristics of Studies Included

First author	Publication year	Sample size	Percentage LVO (%)	Setting	Test administrator	Stroke scales
Almqvist et al.	2021	2,049	12	Pre	EMS/ teleconsulted MD	SSTS
Anadani et al.	2019	439	49	NA	NA	RACE; PASS; FAST-ED; CPSS; 3ISS
Behnke et al.	2021	920	6	Pre	EMS	LAMS
Birnbaum et al.	2019	290	23	Pre	EMS	VAN; NIHSS
Brandler et al.	2020	468	16	Pre	EMS	LAMS
Carbonera et al.	2022	74	41	Pre	EMS	FAST-ED
Carr et al.	2020	478	19	ED	NA	NIHSS; FAST-ED
Carrera et al.	2017	341	21	NA	Neurologist	RACE
Carrera et al.	2019	1,533	20	ED	NA	RACE
Cooray et al.	2018	4,923	41	ED	NA	NIHSS
Crowe et al.	2021	11,319	5	NA	NA	VAN; RACE; LAMS; CPSS
Demeestere et al.	2017	551	25	ED	EMS	NIHSS
Dickson et al.	2019	440	15	Pre	EMS	RACE
Dowbiggin et al.	2022	1,359	11	Pre	EMS	FAST ED; CPSS
Esperon et al.	2022	320	19	ED	Neurologist	Hunter
Ganesh et al.	2022	1,033	12	ED	Paramedics	RACE; PASS; LAMS; G-FAST; FAST PLUS; CPSS; CG-FAST; CSTAT
Gropen et al.	2018	1,663	10	ED	MD	RACE; FAST-ED; EMSA; CSTAT; 3ISS
Gropen et al.	2019	463	10	Pre; ED	Paramedics	NIHSS; FAST-ED; EMSA
Guillory et al.	2019	135	24	Pre; ED	MD	FAST-ED
Heldner et al.	2016	1,085	61	ED	Neurologist	NIHSS
Helwig et al.	2019	53	17	Pre	EMS	LAMS
Hoglund et al.	2020	608	11	ED	MD	FANG
Jumaa et al.	2020	492	37	ED	EMS	RACE
Keenan et al.	2022	184	16	ED	EMS	VAN; SAVE; RACE; PASS; NIHSS; mNIHSS; G-FAST; FAST-ED; CPSS; CSTAT; AAW
Kummer et al.	2016	664	12	ED	Neurologist	CPSS
Lawner et al.	2019	255	50	ED	EMS	RACE; CIPSS
Leibinger et al.	2022	874	59	ED	Neurologist	HEMIPLEGIA
Li et al.	2020	158	4	ED	EMS	CSTAT
Lima et al.	2016	727	33	NA	NA	RACE; NIHSS; FAST-ED; CPSS
Mayasi et al.	2016	274	17	ED	NA	VAN; RACE; FAST-ED; CPSS; 3ISS
Mohamed et al.	2022	907	9	ED	EMS	CPSS
Mulkerin et al.	2021	31	32	Pre	Paramedic	mNIHSS
Noorian et al.	2018	94	48	Pre	Paramedic	VAN; RACE; PASS; NIHSS; LAMS; FAST-ED; 3ISS; CSTAT
Ohta et al.	2019	259	39	ED	MD	RACE; CPSS; FAST-ED; 3ISS
Okuno et al.	2019	502	17	Pre	Paramedic	FACE2AD
Ollikainen et al.	2017	856	13	Pre	NA	PASS; NIHSS; G-FAST; FPSS; FAST-ED; CSTAT; 3ISS

Continued

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Table 1 General Characteristics of Studies Included *(continued)*

First author	Publication year	Sample size	Percentage LVO (%)	Setting	Test administrator	Stroke scales
Ollikainen et al.	2023	302	17	Pre	Paramedic	RACE; MDIRECT; LAMS; FPSS; FAST PLUS; FACE2AD
Ostman et al.	2022	126	19	ED	NA	Hunter; ACT-FAST
Panichpisal et al.	2018	776	12	Pre	NA	VAN; POMONA; PASS; NIHSS; LAMS; CPSS
Panichpisal et al.	2018	1,381	12	Pre	NA	VAN; SNOW; PASS; NIHSS; LAMS; CPSS
Panichpisal et al.	2022	326	10	Pre	Paramedic	SNOW
Pérez de la Ossa et al.	2014	885	9	Pre	EMT	RACE
Puolakka et al.	2022	507	11	Pre	Investigators	VES; SAVE; RACE; POMONA; PASS; LAMS; HEMIPARESIS; G-FAST; FPSS; FAST-ED; FAST PLUS; CSTAT; CPSS; CG-FAST; 3ISS
Purrucker et al.	2017	741	44	ED	Investigators	sNIHSS EMS; sNIHSS; RACE; PASS; FAST-ED; CPSS; 3ISS
Richards et al.	2018	138	43	Pre	Paramedic	CPSS
Suzuki et al.	2018	413	28	Pre	Paramedic	ELVO
Taqi et al.	2019	186	19	Pre	EMS	VES
Teleb et al.	2017	62	23	ED	Nurse	VAN
Thavarajah et al.	2022	126	13	Pre	EMS	RACE
Václavík et al.	2018	435	29	Pre	Paramedic	FAST PLUS
Vidale et al.	2019	145	37	ED	MD; nurse	VAN; NIHSS; LARIO; LAMS; CPSS
Wang et al.	2021	11,440	28	Pre	Paramedic; EMT	VAN; sNIHSS; sNIHSS EMS; ROSIER; RACE; PASS; NIHSS; mNIHSS; LAMS; G-FAST; FPSS; FAST-ED; CPSS; CG-FAST; NIHSS; 4ISS; 3ISS
Wasyliw et al.	2023	1,080	26	Pre	Paramedic	FAST VAN
Yuksen et al.	2023	252	24	ED	MD; paramedic	NIHSS
Zhang et al.	2021	1,053	32	ED	RN	SAFE; RACE; NIHSS; FAST-ED; CPSS; 3ISS
Zhao et al.	2017	565	15	Pre	MD; nurse	RACE; PASS; LAMS; FAST-ED; CPSS
Zhao et al.	2018	104	13	Pre	Paramedic	ACT-FAST
Zhao et al.	2021	517	18	Pre	Paramedic	ACT-FAST

Abbreviations: EMSA = Emergency Medical Stroke Assessment; FAST = Face, Arm, Speech, Time; FAST-ED = Field Assessment Stroke Triage for Emergency Destination; RACE = Rapid Arterial Occlusion Evaluation.

treated as 4 because of the separate analyses conducted on stroke scales in both prehospital and hospital settings. The median number of included patients was 473, with sample sizes ranging from 31 participants to 11,440. The included studies were published between 2014 and 2023, with most (n = 39, 67%) published after 2019. Geographically, most studies were conducted in North America (n = 29, 50%), followed by Europe (n = 15, 26%), Asia (n = 6, 10%), Oceania (n = 6, 10%), and South America (n = 2, 3%) (eFigure 2).

A total of 33 clinical stroke scales were evaluated in this analysis, with the number of studies per scale ranging from 1 to 22. Components of common scales and their relative strengths and weaknesses are listed in eTables 1–2. Considerable variability was observed in sample sizes and LVO prevalence across studies. Among the 58,381 patients included across all studies, 12,277 (21%) were diagnosed with LVO. Regarding study setting, 24 studies (41%) administered stroke scales in a hospital setting, 30 (52%) in a prehospital setting, 2 (3%) in both settings, and 4 (7%) did not specify the setting. Administration of stroke scales also varied by provider type: EMS personnel conducted assessments in 31 studies (53%), physicians in 9 studies (16%), study authors in 2 studies (3%), registered nurses (RNs) in 2 studies (3%), both physicians and RNs in 2 studies (3%), and both physicians and EMS personnel in 1 study (2%). Eleven studies (19%) did not specify who administered the scale. The most commonly used scales were CPSS (17%, n = 10), 3ISS (14%, n = 8), NIHSS (12%, n = 7), and RACE (12%, n = 7). A detailed overview of included studies and stroke scales is presented in Table 1.

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Bivariate Meta-Analysis

Pooled sensitivity and specificity estimates varied substantially across the 33 included stroke scales (Table 2). Rates of LVO per study ranged from 10% to 59%, with all but one study reporting an LVO rate below 40%.

Sensitivity ranged from 0.37 for 3ISS (n = 10 studies, 17,326 participants) to 0.99 for LARIO (n = 1 study, 145 participants). Seven scales had a calculated pooled sensitivity of over 80% (POMONA: n = 2 studies, 1,283 participants; VES: n = 2 studies, 693 participants; FANG: n = 1 study, 608 participants; SNOW: n = 2 studies, 1,707 participants; FACE2AD: n = 2 studies, 804 participants; FAST VAN: n = 1 study, 1,080 participants; and LARIO: n = 1 study, 145 participants). LARIO was the only scale with a reported sensitivity of over 0.86. Specificity ranged from 0.34 for FANG (n = 1 study, 608 participants) to 0.94 for HEMIPLEGIA (n = 1 study, 874 participants) and SSTS (n = 2 studies, 2,905 participants). Six scales had a calculated pooled specificity of over 90% (CG-FAST: n = 3 studies, 12,980 participants; FPSS: n = 4 studies, 13,105 participants; 3ISS: n = 10 studies, 17,326 participants; HEMIPARESIS: n = 1 study, 507 participants; SSTS: n = 1 study, 2,905 participants; and HEMIPLEGIA (n = 1 study, 874 participants).

The summary receiver operating characteristic (ROC) curves for individual scales reveal high diagnostic performance for LARIO (AUC = 0.983), FPSS (AUC = 0.896), FACE2AD (AUC = 0.876), and ACT-FAST (AUC = 0.873) (Figure 1). By contrast, scales such as EMSA (AUC = 0.640) and FANG (AUC = 0.661) showed lower discriminative ability (Table 2). Forest plots for sensitivity and specificity are illustrated in Figure 2, A and B. A subgroup analysis based on clinical setting (prehospital vs hospital) was also performed to summarize the pooled estimates of included scales. In the prehospital subgroup, FPSS (AUC = 0.896), FAST VAN (AUC = 0.878), and FACE2AD (AUC = 0.876) demonstrated robust performance. In the hospital setting, LARIO exhibited the highest diagnostic accuracy (AUC = 0.983), followed by ACT-FAST (AUC = 0.883), 3ISS (AUC = 0.870), and VAN (AUC = 0.870) (Table 3).

A subgroup analysis stratified by test administrator demonstrated that, among EMS personnel, FPSS (AUC = 0.893), ACT-FAST (AUC = 0.889), and FAST VAN (AUC = 0.878) achieved the highest performance. In the physician subgroup, CPSS (AUC = 0.895), 3ISS (AUC = 0.860), and RACE (AUC = 0.852) performed best. For studies in which either EMS personnel or physicians administered the test, the top-performing scales were NIHSS (AUC = 0.954), RACE (AUC = 0.860), and FAST-ED (AUC = 0.852). When tests were administered by either nurses or physicians, the highest performance was observed with LARIO (AUC = 0.983), VAN (AUC = 0.942), and NIHSS (AUC = 0.927). One study using trained emergency department triage nurses to administer the VAN score reported a sensitivity of 0.97 (95% CI: 0.63–1.00), specificity of 0.89 (95% CI: 0.76–0.95), and AUC of 0.975. An overview of these results is presented in eTables 3–4.

Between-study heterogeneity varied widely across stroke scales, with τ^2 values for sensitivity ranging from 0 to 5.70, τ^2 values for specificity ranging from 0 to 3.01, between-study correlations (ρ) ranging from −1.00 to 1.00, and I^2 values spanning 0%–97.7% (eTable 5).

Network Meta-Analysis

A total of 73 stroke scales with thresholds were included in the analysis. Of note, multiple thresholds for the same scale were included to assess intrascale performance in the event that different scale cutoffs were used to define a positive LVO detection in studies included in the final analysis (e.g., NIHSS threshold of 4 points vs NIHSS threshold 8 points and CPSS score of 2 points vs CPSS score of 3 points). An overview of the direct comparisons between stroke scales is illustrated in Figure 3, which shows frequent comparisons for CPSS, NIHSS, and RACE. The network meta-analysis incorporated both direct and indirect comparisons as previously described. eFigure 3 demonstrates the pooled network estimates for diagnostic accuracy. Ranking of SUCRA values identified POMONA (SUCRA = 0.877), NIHSS (0.856), sNIHSS EMS (0.854), G-FAST (0.823), and SAFE (0.788) as the highest-performing tools. These scales demonstrated a favorable balance of sensitivity and specificity. Scales such as ROSIER and EMSA ranked lowest based on SUCRA values. Detailed SUCRA values are given in Table 4. Between-study heterogeneity was τ = 0.49 (95% CrI 0.38–0.61) on the log-DOR scale (τ^2 = 0.24).

Sensitivity Analysis

Funnel plots were produced to illustrate potential publication bias for stroke scales reported by more than one study. Plots for highly studied stroke scales, including RACE, CPSS, NIHSS, and FAST-ED, demonstrated minimal to no publication bias. Scales that were less frequently studied, including VAN, G-FAST, FAST PLUS, Hunter, EMSA, SNOW, and POMONA, demonstrated slight asymmetry in comparison with the more frequently studied scales (eFigures 4–6).

Discussion

Our analysis of 58 studies evaluating the diagnostic accuracy of over 30 clinically based prehospital stroke scales incorporated data from more than 58,000 patients across 18 countries. The LARIO, FPSS, FACE2AD, ACT-FAST, and MDIRECT scales demonstrated the highest overall diagnostic accuracy based on pooled estimates and SUCRA rankings. LARIO exhibited near-perfect sensitivity and specificity in hospital-based studies (AUC = 0.983) while ACT-FAST and 3ISS consistently ranked among the top-performing scales across both prehospital and in-hospital settings. ACT-FAST and MDIRECT also demonstrated strong diagnostic performance, particularly in prehospital environments.

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Table 2 Pooled Test Accuracy for All Stroke Scales

Stroke scale	Studies (patients)	LVO % (n)	Sensitivity (95% CI)	Specificity (95% CI)	PLR (95% CI)	NLR (95% CI)	AUC
3ISS	10 (17,326)	27% (4,650)	0.37 (0.26–0.48)	0.92 (0.90–0.94)	4.76 (3.68–6.01)	0.68 (0.57–0.78)	0.862
4ISS	1 (11,400)	29% (3,244)	0.66 (0.64–0.67)	0.84 (0.83–0.84)	4.03 (3.81–4.26)	0.4 (0.39–0.43)	0.823
ACT-FAST	2 (621)	17% (106)	0.75 (0.59–0.86)	0.85 (0.75–0.92)	5.30 (3.29–8.22)	0.3 (0.18–0.45)	0.873
AAW	1 (184)	16% (29)	0.79 (0.61–0.90)	0.53 (0.45–0.61)	1.68 (1.25–2.11)	0.41 (0.18–0.76)	0.72
CG-FAST	3 (12,980)	26% (3,420)	0.38 (0.17–0.66)	0.91 (0.86–0.95)	4.26 (3.12–4.91)	0.67 (0.40–0.88)	0.866
CPSS	20 (34,166)	18% (6,164)	0.73 (0.65–0.79)	0.77 (0.70–0.83)	3.24 (2.60–4.02)	0.36 (0.28–0.44)	0.815
CSTAT	7 (4,495)	12% (535)	0.54 (0.41–0.67)	0.83 (0.75–0.89)	3.15 (2.40–4.12)	0.56 (0.43–0.68)	0.762
ELVO	1 (413)	28% (114)	0.74 (0.58–0.85)	0.86 (0.67–0.95)	5.92 (2.36–13.4)	0.31 (0.19–0.47)	0.851
EMSA	2 (2,126)	10% (216)	0.59 (0.29–0.83)	0.63 (0.35–0.84)	1.57 (1.28–1.91)	0.66 (0.47–0.85)	0.640
FACE2AD	2 (804)	17% (136)	0.86 (0.79–0.91)	0.79 (0.75–0.82)	4.1 (3.43–4.74)	0.18 (0.12–0.27)	0.876
FANG	1 (608)	11% (64)	0.84 (0.73–0.91)	0.34 (0.30–0.38)	1.27 (1.1–1.42)	0.48 (0.25–0.80)	0.661
FAST PLUS	4 (2,277)	15% (350)	0.73 (0.43–0.91)	0.77 (0.58–0.89)	3.16 (2.10–4.51)	0.36 (0.15–0.64)	0.817
FAST VAN	1 (1,080)	26% (276)	0.86 (0.81–0.90)	0.75 (0.72–0.78)	3.44 (3.02–3.91)	0.19 (0.14–0.25)	0.878
FAST-ED	18 (21,311)	25% (5,308)	0.63 (0.52–0.73)	0.80 (0.71–0.86)	3.10 (2.49–3.81)	0.47 (0.38–0.57)	0.777
FPSS	4 (13,105)	26% (3,458)	0.50 (0.36–0.63)	0.91 (0.89–0.92)	5.54 (3.44–8.10)	0.55 (0.40–0.71)	0.896
G-FAST	5 (14,020)	25% (3,556)	0.71 (0.57–0.81)	0.77 (0.59–0.89)	3.22 (1.94–5.26)	0.39 (0.31–0.49)	0.790
HEMIPARESIS	1 (507)	11% (57)	0.30 (0.19–0.43)	0.92 (0.89–0.94)	3.82 (2.20–6.05)	0.76 (0.62–0.88)	0.741
HEMIPLEGIA	1 (874)	59% (519)	0.45 (0.41–0.49)	0.94 (0.91–0.96)	7.81 (5.01–11.7)	0.58 (0.54–0.63)	0.850
Hunter	2 (446)	19% (86)	0.78 (0.66–0.87)	0.75 (0.53–0.89)	3.34 (1.74–6.42)	0.30 (0.19–0.44)	0.815
LAMS	13 (18,564)	22% (4,050)	0.63 (0.54–0.71)	0.75 (0.66–0.83)	2.58 (1.99–3.35)	0.49 (0.41–0.58)	0.738
LARIO	1 (145)	37% (54)	0.99 (0.87–0.99)	0.82 (0.73–0.89)	5.62 (3.55–8.64)	0.027 (0.001–0.159)	0.983
MDIRECT	1 (302)	17% (50)	0.78 (0.65–0.87)	0.82 (0.77–0.86)	4.41 (3.21–5.89)	0.27 (0.15–0.43)	0.871
mNIHSS	3 (11,655)	25% (3,283)	0.81 (0.71–0.88)	0.56 (0.31–0.78)	1.93 (1.27–3.2)	0.36 (0.33–0.40)	0.786
NIHSS	16 (24,698)	30% (7,424)	0.80 (0.73–0.86)	0.70 (0.63–0.76)	2.67 (2.26–3.14)	0.28 (0.21–0.37)	0.807
PASS	11 (18,016)	25% (4,485)	0.70 (0.57–0.81)	0.78 (0.68–0.86)	3.23 (2.48–4.13)	0.38 (0.28–0.51)	0.809
POMONA	2 (1,283)	12% (151)	0.81 (0.22–0.99)	0.76 (0.42–0.93)	3.14 (1.69–4.5)	0.31 (0.04–0.85)	0.837
RACE	22 (25,076)	24% (5,952)	0.71 (0.62–0.78)	0.73 (0.65–0.80)	2.66 (2.24–3.14)	0.40 (0.33–0.48)	0.780
ROSIER	1 (11,440)	28% (3,244)	0.69 (0.67–0.71)	0.70 (0.69–0.71)	2.27 (2.18–2.37)	0.45 (0.42–0.47)	0.747
SAFE	1 (1,053)	32% (341)	0.69 (0.64–0.73)	0.86 (0.83–0.88)	4.86 (3.98–5.88)	0.37 (0.31–0.43)	0.851
SAVE	2 (691)	12% (86)	0.57 (0.06–0.97)	0.81 (0.30–0.97)	2.69 (1.36–4.42)	0.54 (0.11–0.97)	0.777
sNIHSS	1 (11,440)	28% (3,244)	0.73 (0.67–0.79)	0.73 (0.62–0.82)	2.74 (2.10–3.67)	0.37 (0.34–0.41)	0.785
sNIHSS EMS	2 (12,181)	36% (3,570)	0.67 (0.53–0.79)	0.79 (0.68–0.87)	3.30 (2.32–4.67)	0.42 (0.29–0.56)	0.802
SNOW	2 (1,707)	12% (201)	0.84 (0.73–0.91)	0.59 (0.24–0.87)	2.42 (1.18–5.79)	0.29 (0.20–0.44)	0.834
SSTS	1 (2,905)	11% (316)	0.39 (0.29–0.50)	0.94 (0.88–0.97)	6.41 (3.80–10.30)	0.65 (0.57–0.73)	0.682
VAN	5 (15,152)	25% (3,787)	0.78 (0.67–0.86)	0.68 (0.56–0.79)	2.51 (1.79–3.54)	0.33 (0.21–0.48)	0.796
VES	2 (693)	13% (93)	0.81 (0.24–0.98)	0.84 (0.77–0.89)	4.97 (1.05–9.08)	0.31 (0.02–0.99)	0.863

Abbreviations: EMSA = Emergency Medical Stroke Assessment; FAST = Face, Arm, Speech, Time; FAST-ED = Field Assessment Stroke Triage for Emergency Destination; RACE = Rapid Arterial Occlusion Evaluation.

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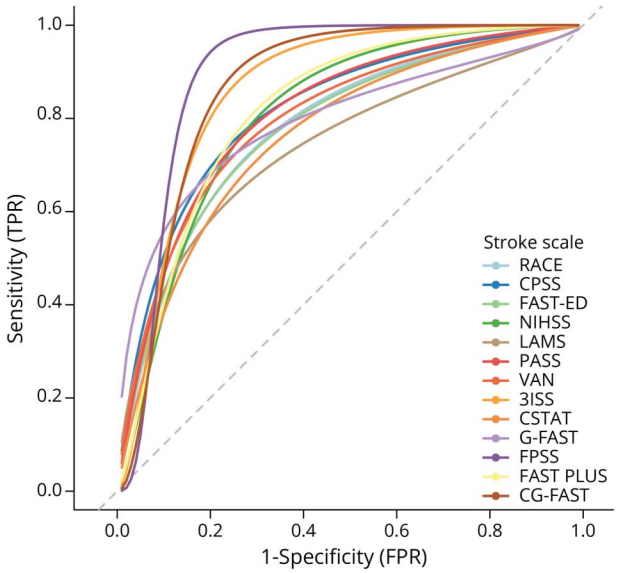
Figure 1 Summary Receiver Operating Characteristic (sROC) Curve Showing the Discriminatory Performance of Stroke Scales



Only stroke scales with more than 3 studies are visualized: RACE, CPSS, FAST-ED, NIHSS, LAMS, PASS, VAN, 3ISS, CSTAT, G-FAST, FPSS, FAST PLUS, and CG-FAST. FAST = Face, Arm, Speech, Time; FAST-ED = Field Assessment Stroke Triage for Emergency Destination; RACE = Rapid Arterial Occlusion Evaluation.

Scales such as FAST-ED, RACE, and NIHSS have been widely accepted as the standard tools for prehospital stroke identification because of their ease of use, clinical familiarity, and inclusion in numerous guideline recommendations and validation studies.[7,25] Although these widely adopted tools showed moderate diagnostic utility in this study, they were outperformed by several less commonly implemented scales. It is important to note that, despite favorable statistical performance among several of these tools, many of the highest-performing scales were limited by factors such as a small number of studies, setting-specific efficacy, or limited external validation. Further research is warranted to validate these findings across diverse patient populations and care environments before a dominant prehospital stroke scale for LVO detection can be established.

Performance of stroke scales has been shown to vary significantly across demographic subgroups. For instance, diagnostic accuracy tends to be higher in younger adults compared with older individuals.[26] Sex-based differences in stroke presentation have also been documented, with women more likely to present with severe strokes and higher rates of LVO than men.[27] These disparities may influence the performance of stroke scales; for example, the RACE scale has demonstrated higher sensitivity but lower specificity in women compared with men.[28] Socioeconomic status may introduce additional variability because individuals from lower socioeconomic backgrounds often present with more severe strokes, potentially affecting the accuracy and generalizability of these diagnostic tools.[29] Together, these findings underscore a fundamental limitation of current clinical stroke scales—their inconsistent performance across diverse patient populations limits their reliability as universal triage tools.

The highest-performing stroke scales in our analysis often demonstrated a tradeoff between sensitivity and specificity. Notably, the top 2 scales for sensitivity (LARIO, FAST VAN) and specificity (SSTS, HEMIPLEGIA) were each evaluated in only a single study, inherently limiting the generalizability and external validity of their results. Although these scales showed incremental improvements over others, the diagnostic performance of physical examination–based stroke scales seems to have plateaued, with sensitivity and specificity typically ranging between 80% and 90%. This ceiling reflects a fundamental limitation of clinical assessment when used in isolation from radiographic or biochemical data. This limitation is further underscored by the continued proliferation of new stroke scales, often tailored to specific geographic or network contexts, yet offering little meaningful innovation over existing tools. The resulting heterogeneity highlights the ongoing challenge of standardizing and implementing a single clinical scale for accurate LVO detection across diverse patient populations and health care settings.

Another critical limitation of existing stroke scales is their inability to reliably distinguish between hemorrhagic and ischemic strokes—an essential differentiation given the vastly different treatment pathways required for each pathology. Misclassification can lead to significant delays in appropriate care, such as prompt blood pressure control and potential neurosurgical intervention for hemorrhagic stroke, vs timely thrombolysis or thrombectomy for ischemic stroke. Inappropriately administering anticoagulants intended for ischemic stroke in the context of a hemorrhagic event could have catastrophic consequences, further underscoring the need for improved diagnostic precision in the prehospital setting.[30-34] Most stroke scales, including those analyzed in this study, are designed to detect the presence and severity of neurologic deficits but lack the pathophysiologic specificity to differentiate between conditions that can cause equally devastating neurologic sequelae. The importance of rapid identification and pathophysiologic specificity is underscored by AHA recommendations of routing patients with suspected LVO to CSCs over non-CSCs and recent studies demonstrating that interhospital transfer of patients with large vessel occlusion stroke to thrombectomy-capable centers is associated with significant delays in initiation of endovascular thrombectomy and is correlated with less favorable neurologic outcomes when compared with direct admission to centers with thrombectomy capability.[9,35-37]

Given these inherent limitations, enhancing diagnostic accuracy requires moving beyond clinical assessments alone, a key step that is actively being addressed through attempts to integrate radiographic and biologic markers into stroke

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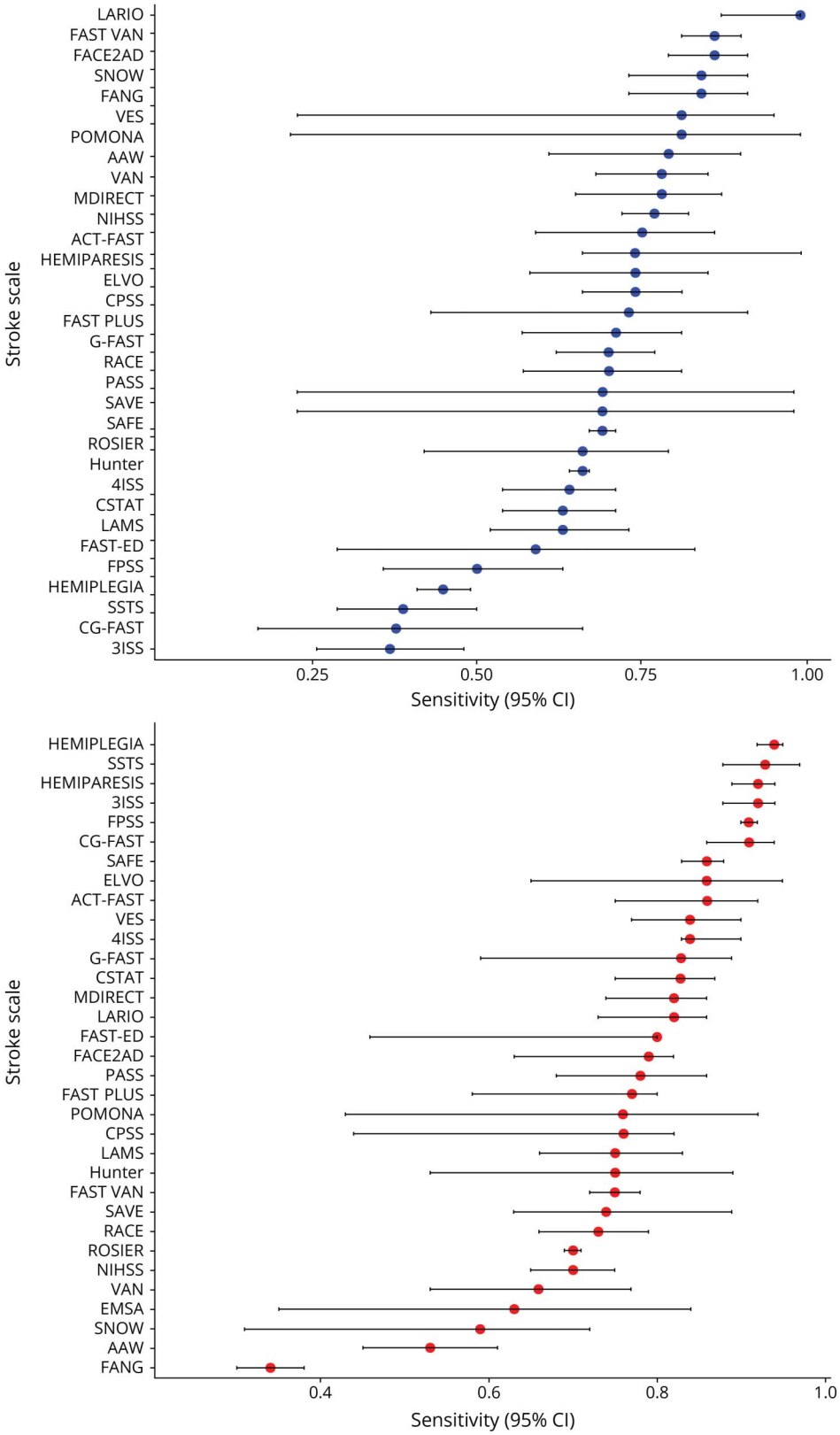
Figure 2 Forest Plots Showing the Sensitivity (A) and Specificity (B) of Various Stroke Scales, With 95%CIs



Panel A (blue) displays sensitivity estimates for each scale while Panel B (red) shows specificity estimates. The black horizontal lines represent the 95% CIs for each study, and the dots indicate the point estimates of sensitivity or specificity. The scales are ordered from lowest to highest sensitivity or specificity.

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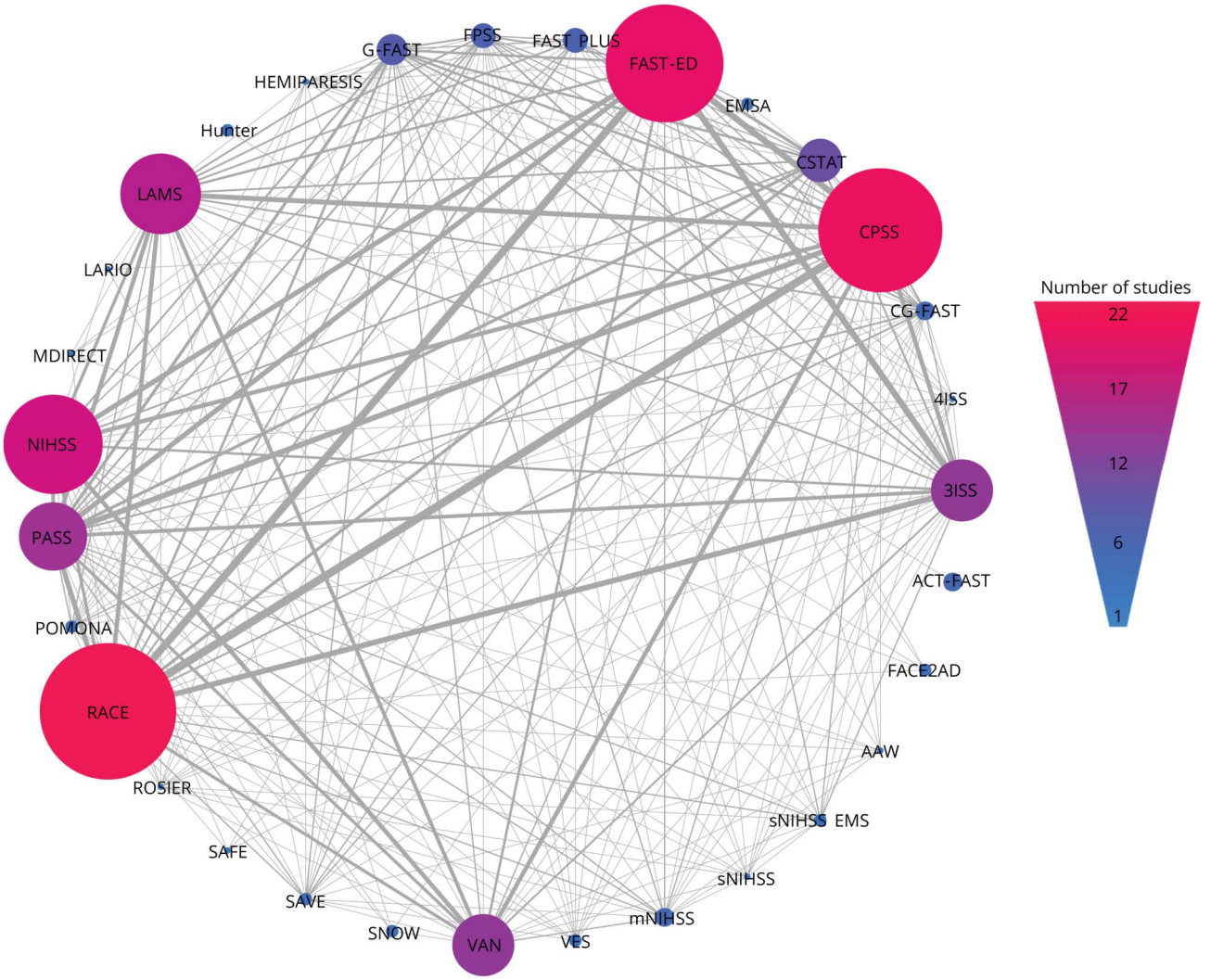
Figure 3 Network Diagram Showing the Relationships Between Various Stroke Scales Based on Direct and Indirect Comparisons



The size of each node represents the number of studies conducted on each scale, with larger nodes indicating more studies. The color gradient (from blue to red) reflects the number of studies, with blue indicating fewer studies and red indicating more studies. The edges between nodes represent direct comparisons, with thicker lines indicating more comparisons between scales. This network visualizes the relative connections and research activity among the included stroke scales.

diagnostics. Studies have assessed the feasibility of mobile stroke units with CT and telemedicine capabilities meant to "bring the hospital to the patient."[38] A recent scoping review discusses the numerous imaging modalities currently being used to detect and stratify stroke with the aim to improve prehospital triage, including ultrasound, near-infrared spectroscopy, and microwave technology, among others.[39] Furthermore, biomarkers offer a promising and objective adjunct to existing stroke scales, with the potential to improve differentiation and early identification of stroke subtypes.[40-42] Recent validation studies have identified glial fibrillary acidic protein (GFAP) and D-dimer as promising biomarkers for differentiating between ischemic and hemorrhagic stroke, underscoring their potential to transform prehospital stroke triage. The Testing for Identification Markers of Stroke trial demonstrated that integrating clinical scales with low GFAP levels (<213 pg/mL) and elevated D-dimer (>600 ng/mL) yielded a sensitivity of 93% and specificity of 80% for detecting LVO.[40] With thoughtful integration into prehospital protocols, these tools could enable first responders to bypass local hospitals in favor of direct transport to TCSs and CSCs, thereby expediting definitive care. Biomarkers also offer a relatively inexpensive and easily implementable diagnostic modality for LVO that may help mitigate limited access to stroke care and stroke imaging resources in low and middle-income countries and geographically restricted regions.[43-46]

The shift from clinical-only stroke diagnosis to the integration of biomarkers mirrors the historical evolution of myocardial infarction (MI) care, where the introduction of cardiac troponins revolutionized diagnostic accuracy. Before the adoption of

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Table 3 Pooled Test Accuracy of Stroke Scales by Test Administration Setting (Prehospital vs In-Hospital)

	Prehospital			In-hospital		
Stroke scale	**Sensitivity (95% CI)**	**Specificity (95% CI)**	**AUC**	**Sensitivity (95% CI)**	**Specificity (95% CI)**	**AUC**
3ISS	0.32 (0.16–0.53)	0.94 (0.90–0.96)	0.873	0.44 (0.29–0.60)	0.92 (0.88–0.94)	0.870
4ISS	0.66 (0.64–0.67)	0.84 (0.83–0.84)	0.823	—	—	—
ACT-FAST	0.83 (0.74–0.89)	0.78 (0.74–0.82)	0.871	0.63 (0.42–0.79)	0.92 (0.85–0.96)	0.883
AAW	—	—	—	0.79 (0.61–0.90)	0.53 (0.45–0.61)	0.724
CG-FAST	0.32 (0.07–0.74)	0.92 (0.83–0.97)	0.866	0.50 (0.41–0.59)	0.89 (0.87–0.91)	0.802
CPSS	0.77 (0.51–0.92)	0.61 (0.26–0.88)	0.764	0.73 (0.64–0.81)	0.82 (0.76–0.87)	0.850
CSTAT	0.51 (0.28–0.74)	0.88 (0.76–0.94)	0.810	0.61 (0.46–0.74)	0.77 (0.65–0.86)	0.739
ELVO	0.74 (0.58–0.85)	0.86 (0.67–0.95)	0.851	—	—	—
EMSA	—	—	—	0.59 (0.29–0.83)	0.37 (0.16–0.65)	0.640
FACE2AD	0.86 (0.79–0.91)	0.79 (0.75–0.82)	0.876	—	—	—
FANG	—	—	—	0.84 (0.73–0.91)	0.34 (0.30–0.38)	0.661
FAST PLUS	0.78 (0.38–0.95)	0.74 (0.46–0.90)	0.818	0.60 (0.51–0.68)	0.84 (0.81–0.86)	0.798
FAST VAN	0.86 (0.81–0.90)	0.75 (0.72–0.78)	0.878	—	—	—
FAST-ED	0.69 (0.49–0.84)	0.72 (0.54–0.85)	0.761	0.65 (0.50–0.77)	0.81 (0.69–0.89)	0.792
FPSS	0.50 (0.36–0.63)	0.91 (0.89–0.92)	0.896	—	—	—
G-FAST	0.63 (0.45–0.78)	0.85 (0.64–0.94)	0.789	0.82 (0.62–0.93)	0.64 (0.36–0.84)	0.809
HEMIPARESIS	0.30 (0.19–0.43)	0.92 (0.89–0.94)	0.741	—	—	—
HEMIPLEGIA	—	—	—	0.45 (0.41–0.49)	0.94 (0.91–0.96)	0.85
Hunter	—	—	—	0.78 (0.66–0.87)	0.75 (0.53–0.89)	0.815
LAMS	0.71 (0.56–0.83)	0.75 (0.62–0.85)	0.796	0.48 (0.21–0.76)	0.73 (0.38–0.92)	0.629
LARIO	—	—	—	0.99 (0.87–0.99)	0.92 (0.73–0.89)	0.983
MDIRECT	0.78 (0.65–0.88)	0.82 (0.77–0.86)	0.871	—	—	—
mNIHSS	0.68 (0.57–0.78)	0.72 (0.57–0.83)	0.750	0.93 (0.45–0.99)	0.38 (0.13–0.72)	0.679
NIHSS	0.86 (0.71–0.94)	0.63 (0.51–0.74)	0.785	0.78 (0.71–0.83)	0.75 (0.67–0.81)	0.826
PASS	0.62 (0.41–0.80)	0.83 (0.75–0.89)	0.826	0.71 (0.53–0.84)	0.27 (0.46–0.90)	0.770
POMONA	0.81 (0.22–0.99)	0.76 (0.42–0.93)	0.837	—	—	—
RACE	0.71 (0.58–0.81)	0.70 (0.57–0.80)	0.763	0.77 (0.61–0.88)	0.71 (0.55–0.83)	0.801
ROSIER	0.69 (0.67–0.71)	0.70 (0.69–0.71)	0.747	—	—	—
SAFE	—	—	—	0.69 (0.64–0.73)	0.86 (0.83–0.88)	0.851
SAVE	0.23 (0.14–0.35)	0.93 (0.90–0.95)	0.709	0.86 (0.69–0.95)	0.57 (0.49–0.64)	0.804
SNOW	0.84 (0.73–0.91)	0.59 (0.24–0.87)	0.834	—	—	—
sNIHSS	0.73 (0.67–0.79)	0.73 (0.62–0.82)	0.785	—	—	—
sNIHSS EMS	0.82 (0.80–0.83)	0.63 (0.62–0.64)	0.797	0.61 (0.49–0.73)	0.83 (0.78–0.88)	0.825
SSTS	0.39 (0.29–0.50)	0.94 (0.88–0.97)	0.682	—	—	—
VAN	0.76 (0.55–0.89)	0.64 (0.44–0.81)	0.758	0.83 (0.69–0.91)	0.77 (0.59–0.89)	0.870
VES	0.81 (0.24–0.98)	0.84 (0.77–0.89)	0.863	—	—	—

Abbreviations: EMSA = Emergency Medical Stroke Assessment; FAST = Face, Arm, Speech, Time; FAST-ED = Field Assessment Stroke Triage for Emergency Destination; RACE = Rapid Arterial Occlusion Evaluation.

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Table 4 Ranking of Stroke Scales With Threshold Values Based on SUCRA Analysis

Stroke scale	Threshold	SUCRA	Rank	2.5th quantile rank	50th quantile rank	97.5th quantile rank
POMONA	1	0.873	1	1	4	61
NIHSS	8	0.855	2	1	4	62
sNIHSS EMS	8	0.853	3	1	6	53
G-FAST	4	0.821	4	1	8	58
mNIHSS	2	0.788	5	1	3	72
SAFE	4	0.787	6	2	11	55
PASS	3	0.741	7	1	11	66
CPSS	1	0.738	8	4	17	50
NIHSS	10	0.73	9	6	18	48
4ISS	4	0.727	10	3	16	57
CSTAT	4	0.685	11	2	17	65
NIHSS	4	0.66	12	1	13	72
G-FAST	3	0.656	13	7	24	53
FAST-ED	4	0.654	14	12	26	45
RACE	5	0.653	15	12	26	44
CPSS	3	0.639	16	9	26	53
SAVE	2	0.638	17	2	19	70
FAST-ED	5	0.632	18	7	26	56
3ISS	3	0.63	19	3	23	67
CPSS	2	0.622	20	14	28	46
mNIHSS	7	0.616	21	5	26	63
sNIHSS	6	0.615	22	5	27	63
FAST-ED	6	0.612	23	8	28	57
RACE	3	0.597	24	10	30	54
sNIHSS	4	0.594	25	5	29	63
RACE	2	0.594	26	7	30	60
NIHSS	12	0.589	27	5	29	64
G-FAST	2	0.58	28	1	26	72
RACE	4	0.578	29	9	31	57
CG-FAST	4	0.577	30	7	31	62
sNIHSS EMS	6	0.569	31	6	31	64
NIHSS	6	0.555	32	12	33	56
FAST-ED	2	0.551	33	8	34	62
FPSS	5	0.548	34	6	34	65
VES	1	0.525	35	4	36	69
POMONA	2	0.504	36	7	38	67
RACE	7	0.502	37	14	38	59
PASS	2	0.5	38	16	38	57

Continued

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Table 4 Ranking of Stroke Scales With Threshold Values Based on SUCRA Analysis *(continued)*

Stroke scale	Threshold	SUCRA	Rank	2.5th quantile rank	50th quantile rank	97.5th quantile rank
FAST-ED	7	0.488	39	11	39	63
RACE	6	0.485	40	14	39	61
FAST-ED	8	0.48	41	8	40	66
FAST PLUS	2	0.457	42	5	44	70
FAST-ED	3	0.455	43	18	42	60
FAST-ED	1	0.441	44	8	44	69
RACE	1	0.44	45	10	44	67
RACE	9	0.436	46	13	44	65
FPSS	4	0.425	47	5	47	71
RACE	8	0.417	48	17	45	64
EMSA	6	0.409	49	4	50	72
mNIHSS	6	0.409	50	4	50	72
LAMS	5	0.406	51	7	48	70
3ISS	4	0.396	52	15	45	70
SAVE	3	0.386	53	6	52	72
sNIHSS	2	0.384	54	11	49	69
VAN	2	0.378	55	11	50	69
LAMS	1	0.377	56	7	52	71
FAST-ED	9	0.37	57	10	51	70
LAMS	3	0.367	58	8	52	71
CSTAT	2	0.358	59	13	52	69
NIHSS	7	0.354	60	5	56	72
NIHSS	9	0.347	61	7	55	72
PASS	1	0.338	62	3	60	73
LAMS	2	0.336	63	8	55	71
LAMS	4	0.33	64	29	51	64
EMSA	3	0.322	65	10	56	71
EMSA	2	0.3	66	11	58	71
ROSIER	4	0.296	67	16	56	70
sNIHSS EMS	5	0.274	68	10	60	72
EMSA	4	0.255	69	15	60	72
EMSA	1	0.193	70	20	65	72
NIHSS	2	0.159	71	35	65	72
EMSA	5	0.095	72	45	70	73
FAST-ED	0	0.038	73	22	74	74

Abbreviations: EMSA = Emergency Medical Stroke Assessment; FAST = Face, Arm, Speech, Time; FAST-ED = Field Assessment Stroke Triage for Emergency Destination; RACE = Rapid Arterial Occlusion Evaluation.

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biomarkers, MI was diagnosed primarily through clinical assessment and electrocardiogram (ECG) findings. This included the evaluation of symptoms such as chest pain, vital sign abnormalities, and physical examination findings suggestive of cardiac compromise, which naturally lacked both sensitivity and specificity, especially in atypical presentations.[47] This changed with the recognition that MI could be reliably identified by a characteristic rise and/or fall in cardio-specific troponin I or T levels.[48] Today, troponin measurement remains the gold standard for MI diagnosis. Stroke care now stands at a similar inflection point, with biomarker integration offering the potential for a comparable transformation in diagnostic precision and timely intervention.

This study has several limitations to be acknowledged. First, the analysis was restricted to studies involving patients with LVO, excluding those focused on other stroke subtypes. In addition, included studies encompassed diverse clinical settings and were conducted by personnel with varying levels of medical training across a wide range of health care systems and geographic regions—each with distinct protocols and standards of care. This heterogeneity introduces challenges to both internal and external validity. Furthermore, while several studies have shown that both physicians and nonphysicians (e.g., EMTs, nurses, and physician assistants) can perform stroke scale assessments with comparable accuracy, we were unable to conduct a comprehensive analysis of outcome differences among nonphysician staff trained specifically in stroke scale administration, due to limited data available in the included studies.[49-51] Finally, the wide variety of stroke scales assessed complicates direct comparisons across studies.

In conclusion, while clinical stroke scales demonstrate reasonable overall diagnostic accuracy, some novel tools seem to outperform more widely adopted scales. Nevertheless, the sensitivity and specificity of physical examination–based tools alone for detecting LVO seem to have reached a performance plateau. To fully optimize prehospital stroke triage and improve outcomes in the era of mechanical thrombectomy, the integration of objective biomarkers and/or portable tools will be essential.

Author Contributions

N.L.A. Nawabi: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. E.R. Rivera: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. D. de Wilde: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J.L. Kilgallon: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design. D.I. Nawabi: major role in the acquisition of data; analysis or interpretation of data. V.M. Bhave: major role in the acquisition of data; study concept or design. P. Emedom-Nnamdi: analysis or interpretation of data. A. Kappel: study concept or design. S.D. Rangwala: study concept or design. R.E. Alcedo Guardia: study concept or design. J. Vicenty-Padilla: study concept or design. S. Izzy: study concept or design. N.J. Patel: study concept or design. R Du: drafting/revision of the manuscript for content, including medical writing for content. A.A. Dmytriw: drafting/revision of the manuscript for content, including medical writing for content. T. Gropen: study concept or design. D.S. Liebeskind: study concept or design. E. Gaude: analysis or interpretation of data. A.P. See: study concept or design; analysis or interpretation of data. M.A. Aziz-Sultan: study concept or design; analysis or interpretation of data. J.D. Bernstock: study concept or design; analysis or interpretation of data.

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Disclosure

E. Gaude is a shareholder of Upfront diagnostics. J.D. Bernstock has an equity position in Treovir Inc. and UpFront Diagnostics. J.D. Bernstock is also on the Synaptive, Centile Bioscience, QV Bioelectronics, and NeuroX1 boards of scientific advisors. The remaining authors declare that the article and its contents were composed in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. Go to Neurology.org/N for full disclosures.

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